A Signed-Winding Dichotomy for Canonical Real-Time Pushdown Realization under Periodic Height Guards

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Abstract

Consider two independent, fully tagged Dyck-prefix stacks whose input letters perform one push or matching pop on exactly one source stack. Impose the additional all-prefix height constraint consisting of both free coordinate axes and an arbitrary rectangularly periodic mask in the positive quadrant. We give a finite signed-winding criterion for the resulting empty-to-empty language to be context-free. In the positive cases, a single deterministic physical stack preserves the two complete ordered source contents in a history-independent canonical representation, reflecting every legal and illegal source transition on the *same* input letter without epsilon moves. In the negative cases, an accessible rank-two voltage component supplies a crossed-equality obstruction and a negatively sloped rank-one component supplies a three-equal-block obstruction. The finite graph criterion is computable in linear word-RAM time in the size of an explicit mask.

Keywords: deterministic pushdown automata; Dyck languages; periodic graphs; voltage subgroup; source-faithful realization; real-time compilation; non-context-free languages.
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1 Introduction

Combining two independent pushdown computations does not in general preserve context-freeness: their two unrelated last-in–first-out obligations can enforce crossed equalities. Extra restrictions on the heights of the two stacks can make one physical stack sufficient, but the answer depends on the *paths of legal prefixes*, not just the dimensions of the ambient region. We study a natural explicitly periodic class in which this dependence can be stated exactly.

The main theorem gives an exact criterion for context-freeness of the language recognized by the guarded pair of tagged Dyck stacks. It also identifies precisely when the entire source transition system has a rooted, history-independent and full-source-injective physical realization by one deterministic real-time pushdown automaton. The equivalence is stronger than language recognition alone: the physical representation preserves the ordered contents of both sources and reflects the legality of every individual input action.

Periodic graph walks and counter machines have established connections [1]; general semilinear descriptions of languages have a classical foundation [2]. Earlier work on multistack models, including bounded balance and scope restrictions, provides important context [3, 4]. Real-time strict deterministic languages are also well established [5]. More recent results concerning regular-trajectory shuffles and geometric obstructions for context-free intersections provide complementary perspectives [6, 7]. The present classification concerns a different, explicit object: all-prefix guards determined by the two live source-stack heights.

The distinction between an abstract semantic state and the information retained by a physical pushdown representation is useful in the broader normalization setting [8, 9]. Here the complete encoding and both non-context-freeness arguments are provided directly, without relying on any unavailable supporting document. In particular, the positive construction and the negative regular-language witnesses are proved for each allowed or forbidden signed-winding type.

Main contributions

(i) A signed-winding criterion, valid for every explicitly given rectangular periodic height mask and all periods, that characterizes context-freeness in the stated fully tagged two-source Dyck model.

(ii) A history-independent, rooted, full-source one-stack construction for every permitted winding pattern, with exact typed-pop rejection and one input-consuming transition per source letter.

(iii) Two complementary impossibility certificates: rank-two components yield crossed equalities, whereas oppositely directed rank-one winding yields a three-equal-block obstruction.

(iv) An effective linear-time test for explicit masks, together with non-affine examples and sharp reachable-height and physical-information bounds.

The finite-graph method, buffering, and block acceleration are used as established tools; the contribution is their exact necessity–sufficiency synthesis under the stated source/action contract. This is not a statement about arbitrary Presburger guards, simultaneous source-stack actions, epsilon-transition source machines, or succinct mask encodings.

2 The guarded two-source model and its finite quotient

Fix integers $p, q \geq 1$, two nonempty *finite* tag alphabets T_X, T_Y, and a residue mask

$$R \subseteq \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}.$$

Let X_t^+, X_t^- and Y_s^+, Y_s^- denote tagged push and matching-pop input letters ($t \in T_X$, $s \in T_Y$). A source configuration is a pair $(U, V) \in T_X^* \times T_Y^*$, written bottom-to-top. A push appends its tag; a pop is defined exactly when the source top has the prescribed tag. Each input letter changes *one* of U, V, and never the other.

The height safety set is

$$S_R = \{(x, y) \in \mathbb{N}^2 : xy = 0\} \ \cup \ \{(x, y) \in \mathbb{N}^2 : x, y \geq 1, \ (x \bmod p, y \bmod q) \in R\}.$$

A word belongs to L_R if every prefix has defined source-stack action and a height in S_R, and the terminal configuration is $(\varepsilon, \varepsilon)$. Write $\mathcal{C}_R$ for the full *root-reachable* source transition system; preserving only heights would not preserve its tags.

Define the signed residue *multigraph* G_R: its vertex set is R, and for each $r \in R$ and each $e \in \{\pm(1, 0), \pm(0, 1)\}$ it has an arc $r \xrightarrow{e} r + e$ when $r + e \in R$ modulo (p, q). Parallel arcs, including $+1$ and -1 arcs with the same endpoints when a period is 1 or 2, are distinct. Every arc has its integer displacement e. For a connected component K, its *voltage subgroup* $H_K \leq p\mathbb{Z} \times q\mathbb{Z}$ consists of all displacements of closed quotient walks based at a fixed vertex. It is independent of the basepoint and has rank $0, 1$ or 2.

Choose a finite spanning tree $\mathcal{T}_K$ and integer potentials $t(r) = (t_x(r), t_y(r))$ obtained by summing the edge labels along tree paths from a root with potential zero. The vector

$$\omega(r \xrightarrow{e} s) = t(r) + e - t(s)$$

is a cycle voltage; the nonzero such vectors generate H_K. If $N = \max(1, |R|)$, tree potentials have coordinate absolute values at most N, and chord voltages at most $2N + 1$ in absolute value.

Theorem 2.1 (Signed-winding realization dichotomy)**.** *For the model above, the following statements are equivalent.*

(i) For every connected component K of G_R, either $\operatorname{rank} H_K = 0$*, or* $\operatorname{rank} H_K = 1$ *with a nonzero generator (A_K, B_K) satisfying $A_K B_K \geq 0$.*

(ii) The reachable guarded source system $\mathcal{C}_R$ has an effective, rooted, history-independent, full-source-injective, transition-equivariant, domain-reflecting, same-input, epsilon-free real-time one-physical-stack representation. Every input letter requires at most one physical top-stack replacement.

(iii) $L_R \in \mathrm{DCFL}$.

(iv) $L_R \in \mathrm{CFL}$.

If these conditions hold, a finite graph certificate and the resulting physical machine can be constructed effectively. If they fail, an explicit regular-macro non-context-free witness can be extracted from an offending component.

The ordinary logical implications (ii)⇒(iii)⇒(iv) are immediate. Sections 3 and 4 establish the substantive implications (iv)⇒(i) and (i)⇒(ii).

3 Two unconditional negative obstructions

Here the argument can be restricted to a *single tag* in each source, since a regular restriction of a context-free language must remain context-free. Replace each legal unit-step word W by its reverse *inverse* $\overline{W}$, obtained by reversing its letter sequence and exchanging the corresponding pushes and pops. When lifted from the endpoint of W, $\overline{W}$ retraces the path.

Lemma 3.1 (Automatic accessibility of a positive axis loop)**.** *Let K contain a based closed quotient walk Q with net displacement $(0, D)$, $D > 0$. Its repeated lift can be placed sufficiently far to the right so that a positive suffix begins at a vertex $(x_e, 1)$ and stays in $\mathbb{N}^2_{>0}$, reaching a lift of the chosen basepoint at arbitrarily large height. The suffix is reachable from the empty source by a free-X-axis prefix followed by one Y push.*

Proof. The bi-infinite lift $Q^{\mathbb{Z}}$ has bounded horizontal deviation and tends to opposite infinities vertically in its two time directions. Translate it horizontally by a sufficiently large multiple of p so every horizontal coordinate is positive. There are only finitely many visits to height 1 at nonnegative time; choose the *last* such visit after fixing an arbitrary origin of time. Every later vertex has $y \geq 1$, since any later $y = 0$ excursion would recross $y = 1$. A full turn reaches the basepoint; additional turns raise the height by D. Travel on the free X axis to $(x_e, 0)$ and push a Y symbol to enter at $(x_e, 1)$. The remaining fixed suffix is legal. □

Theorem 3.2 (Rank two excludes every PDA)**.** *If some H_K has rank two then* $L_R \notin \mathrm{CFL}$.

Proof. Let U, V be based closed quotient words with independent net displacements $g = (g_x, g_y)$, $h = (h_x, h_y)$. By concatenation and inversion, integer linear combinations of these based loops yield based quotient words P, Q with displacements $(D, 0)$ and $(0, D)$, respectively, for $D = |\det(g, h)| > 0$. Since $g, h \in p\mathbb{Z} \times q\mathbb{Z}$, D is divisible by both periods. Thus every translation by $i(D, 0) + j(0, D)$ preserves allowed interior residues.

Lemma 3.1 applied to Q supplies a legal entry word u from the root to a lift z of the base vertex. Choose z sufficiently deep and far right that the finite paths P, Q and their reversals, based there, remain in the strictly positive quadrant. Then for every $m, n \geq 0$ the word

$$uP^m Q^n \overline{P}^m \overline{Q}^n \overline{u}$$

(where exponentiation applies to the full inverse words) is accepted: its pieces are legal nonnegative period translates of the anchored paths, and it returns to the root. Conversely a word of the regular macro shape

$$uP^m Q^n \overline{P}^k \overline{Q}^\ell \overline{u}$$

can be accepted only if its terminal height is zero, which implies $(m-k)D = (n-\ell)D = 0$. Hence the macro exponent language is exactly

$$\{\alpha^m \beta^n \gamma^m \delta^n : m, n \geq 0\}.$$

The inverse homomorphic preimage of L_R under the four fixed macros, after taking fixed left/right contexts and intersecting with the regular block expression, is this standard non-CFL. Closure of CFLs under these operations yields the contradiction. □

Theorem 3.3 (Opposite rank-one winding excludes every PDA)**.** *If some component has a rank-one voltage direction with negative coordinate product, then* $L_R \notin$ CFL*.*

Proof. Reverse an appropriate based closed quotient walk if needed to get net displacement $(-A, B)$ with $A, B > 0$ and $p \mid A$, $q \mid B$. Lift its base z deep into the positive quadrant. Repeating the inverse walk backward decreases y and increases x without bound; the first hit of $y = 1$ on this backward trajectory provides a fixed positive path U from $(x_e, 0)$, after one Y push, to z. Repeating the original walk forward decreases x and increases y without bound; the first hit of $x = 1$ gives a positive exit path V to $(0, y_f)$, after one X pop. The finite entry and exit paths may be translated by horizontal multiples of A and vertical multiples of B.

Choose fixed source words u for the entry after a^{x_e} and v for the exit to the Y axis. Consider the regular macro family

$$a^{x_e} (a^A)^m \, u \, T^n \, v \, (C^B)^\ell C^{y_f}, \qquad m, n, \ell \geq 0.$$

Here a pushes X, C pops Y, and the fixed u, v contain all necessary boundary steps. Terminal X and Y heights are respectively $A(m-n)$ and $B(n-\ell)$; therefore acceptance forces $m = n = \ell$. If they are equal, the T loops traverse translates indexed by $(A(m-1-j), Bj)$, and the translated entry/exit paths are allowed; the final Y pops occur on the free axis. Thus all and only matched triple exponents are accepted. Fixed-context rational transduction would make $\{\alpha^m \beta^m \gamma^m : m \geq 0\}$ context-free if L_R were context-free, a contradiction. □

4 The canonical positive construction

Assume now that every voltage component passes the sign/rank test. All source tags and their original order must be recoverable, not merely their heights. We first define a *logical* source-labelled word $W(U, V)$ and finite auxiliary tails; then prove uniform top locality and compile it into a physical word using one stack edit per input.

Set $N = \max(1, |R|)$, $D = 2N + 1$, $M = 4N^2 + 20$, and

$$E = M + 2N^2 + 2N + 1, \qquad C = 6N^2 + 7N + 70. \tag{1}$$

These are intentionally generous explicit constants, independent of source heights or tag contents.

4.1 Rank-zero lifted islands and short-source modes

Lemma 4.1 (Sharp bounded-minimum lemma)**.** *In any root-reachable rank-zero interior lifted component* K*,* $\min(x, y) \leq |K| \leq N$*.*

Proof. A zero-voltage lift contains exactly one vertex for each quotient residue in K, hence exactly $|K|$ vertices and graph diameter at most $|K| - 1$. The last positive-quadrant segment of a rooted path enters at $(x_e, 1)$ or $(1, y_e)$. Every vertex of that segment is at graph distance at most $|K| - 1$ from its entry, so respectively $y \leq |K|$ or $x \leq |K|$. □

Define the rank-zero normal form Z_F when $|U|, |V| \le N$ by storing both complete strings in finite control and setting $W = \varepsilon$. In Z_X when $|U| > N \ge |V|$, let $W = U$ and store V in control; in Z_Y, exchange roles. Mode and bounded stack tails are uniquely determined by the full source configuration. Any threshold crossing at N changes a top suffix of length at most $N + 1$, and no one-source-letter step can jump directly from Z_X to Z_Y.

4.2 Axis-parallel rank-one components

For a horizontal-winding component, choose tree potential t, and set

$$L_H(r) = N + 4 + t_y(r) - \min_{s \in K} t_y(s), \qquad b_Y = \max(0, |V| - L_H(r)).$$

Define $W = V[: b_Y]\, U$ and cache $V[b_Y :]$ in finite control. The latter tail has length at most $2N + 3$. Every interior edge $r \xrightarrow{(d_x, d_y)} s$ satisfies $t_y(s) = t_y(r) + d_y$, since all cycle voltages have zero vertical coordinate. Therefore b_Y is an exact interior invariant: changing X touches the exposed U top, and changing Y touches the cached Y tail only.

For a vertical-winding component symmetrically put

$$L_V(r) = N + 4 + t_x(r) - \min_{s \in K} t_x(s), \quad b_X = \max(0, |U| - L_V(r)),$$

and take $W = U[: b_X]\, V$, caching the bounded remaining X suffix. On an axis crossing, the old axis word and new interior word agree below all but at most $2N + 3$ source cells, since the frozen base is zero at the incident short-coordinate axis, or all but a bounded transverse tail at the opposite axis.

4.3 Same-sign rank-one components: aligned tagged frames

Let the primitive positive winding direction of K be (A, B), $A, B \ge 1$. All fundamental cycle voltages are integer multiples of this vector. It follows that $A, B \le D$. For $r = (x \bmod p, y \bmod q)$ define the *transverse invariant*

$$\chi = B(x - t_x(r)) - A(y - t_y(r)). \tag{2}$$

Every allowed interior edge preserves χ: the failure of $t(s) - t(r)$ to equal the edge displacement is a multiple of (A, B).

If $\chi \ge 0$, set

$$b_X = \max\{0, \lfloor \chi / B \rfloor - M\}, \quad b_Y = 0;$$

if $\chi < 0$, put

$$b_X = 0, \quad b_Y = \max\{0, \lfloor -\chi / A \rfloor - M\}.$$

Both base lengths are nonnegative, do not exceed their source heights, and remain constant on every interior step. Write

$$n_f = \min\left\{ \left\lfloor \frac{x - b_X}{A} \right\rfloor, \left\lfloor \frac{y - b_Y}{B} \right\rfloor \right\}.$$

The logical word starts with the first b_X X-tags or b_Y Y-tags, labelled by their source. Above this *frozen base*, it has n_f ordered *paired frames*, frame i containing the next A X-tags and B Y-tags. Store the remaining top tails

$$U[b_X + n_f A :], \qquad V[b_Y + n_f B :]$$

in finite control. The first frame has a special marker; the remaining frames have an ordinary marker. The first original source cell of each nonempty stack likewise carries a finite depth-one marker wherever it resides. These finite marks make exact source emptiness and singleton-stack detection possible without scanning the physical stack.

Lemma 4.2 (Bounded tagged tails)**.** *The Northeast normal form above is defined for every positive source pair with residue in K; each finite cached tail has length at most $E + D + 1 < C$. The frozen base is invariant under interior edges, and n_f changes by at most one under any such edge.*

Proof. It suffices to analyze $\chi \geq 0$; the other case exchanges X and Y. Put $e = (x - b_X) - (A/B)y$. Since $|t_x|, |t_y| \leq N$ and $A, B \leq D$, if $b_X > 0$, then

$$e = M + t_x - (A/B)t_y + \{\chi/B\} > M - N - DN > 0,$$

and $e \leq E$. Hence $n_f = \lfloor y/B \rfloor$; the remaining Y suffix has length less than B, and the remaining X suffix has length less than $E + A$. When $b_X = 0$, we have $\lfloor \chi/B \rfloor \leq M$, so $-(N + DN) \leq e \leq E$. If X becomes the limiting source, its remainder is $< A$ and the remaining Y suffix is at most $B + (B/A)(N + (A/B)N) \leq D + DN + N$. Otherwise the previous bounds apply. All are smaller than $E + D + 1$. Invariance of b_X follows from invariance of χ; one source height changes by one on an interior step, so a floor quotient in the definition of n_f changes by at most one. Finally $M > N + DN$ prevents a nonempty frozen base from being the source top: above such a base there is a positive, mask-bounded-margin portion. □

4.4 Free-axis interfaces, top locality and one-stack packing

For an axis configuration (U, ε) or (ε, V) take W to be its sole nonempty tagged source word and store the usual finite mode/residue flags. A transition to an interior rank-zero component is top-local by Lemma 4.1. The horizontal/vertical case follows from the frozen-base formula. The remaining axis interface is explicit:

Lemma 4.3 (Northeast entry from either free axis)**.** *At a legal X-axis entry $y : 0 \to 1$ into a northeast component, with $r = (x \bmod p, 1 \bmod q)$ and $t(r) = (t_x, t_y)$, the new frozen X-prefix when $\chi \geq 0$ is*

$$b_X = \max\{0, x - k_X(r)\}, \qquad k_X(r) = M + t_x + \left\lceil \frac{A(1 - t_y)}{B} \right\rceil.$$

If $\chi < 0$, then the old X height is bounded by a mask-dependent constant. The reverse exit is top-local. The statements with X and Y interchanged hold on the Y axis.

Proof. The displayed formula follows directly from $\lfloor \chi/B \rfloor = x - t_x - \lceil A(1 - t_y)/B \rceil$. If $\chi < 0$, then $x < t_x + (A/B)(1 - t_y)$, which is bounded solely in N, D. In the active positive branch all but $k_X(r)$ cells of the axis X word form an unchanged physical prefix; if inactive, the entire old word has bounded length. The new logical word contains at most one paired frame since $y = 1$. All content to be rearranged is therefore in a fixed-length top suffix. A reverse crossing has $y = 1$ and removes its lone Y source tag. The first-paired-frame marker, source-bottom tag markers and bounded caches distinguish this event from a deeper pop, even when a lower physical block is hidden. No complete source word is consulted. The Y-axis proof is symmetric. □

Lemma 4.4 (Finite top-local transduction)**.** *For all root-reachable source states satisfying the signed-winding condition, every defined or undefined one-letter source action can be decided from finite control and a bounded logical top suffix. On a defined action, the logical normal forms before and after have a common prefix and each changed suffix has length at most C in* (1)*. The necessary replacement suffix and new finite control are computable from those top data alone.*

Proof. Interior-to-interior edges remain in the same quotient component by construction of G_R. In rank-zero modes, bounded minimum height and threshold flags suffice. In horizontal/vertical modes, the transverse frozen base is unchanged and the other source is fully exposed above it, while the transverse source's top is in the bounded cache. In northeast modes the frozen base and its direction are invariant, and only the top paired frame and the two bounded tails may change; Lemma 4.2 bounds their sizes. Source top-tag matching is therefore visible. On an axis boundary, Lemma 4.3 or its rank-zero/axis-parallel analog gives a uniformly bounded common-prefix rewrite and a finite threshold test depending only on the residue and visible source-bottom/first-frame marks. The constants N, D, M, E, C were selected to dominate all individual cache and axis rewrite bounds (the largest is $O(N^2)$). Rejection is detected before changing the physical configuration. Thus a finite transition table over fixed finite cached strings and logical suffixes exists. □

Lemma 4.5 (One-operation block packing)**.** *Suppose a logical word W has a top-local transition that replaces a suffix of length at most C by one of length at most C. There is a unique fixed-block representation with a finite top buffer and a physical stack of C-cell blocks in which every logical input transition uses at most one physical stack push or pop, or a no-op, and no epsilon move.*

Proof. Store all of W in finite control if $|W| < 2C$. Otherwise place as many full C-cell blocks as possible on the physical stack while keeping exactly $C, \dots, 2C-1$ remaining top cells in finite control. The canonical choice is $k = \lfloor (|W| - C)/C \rfloor$ blocks; then $|W| - kC \in [C, 2C-1]$. The last C old cells lie inside the top buffer, so the source operation is evaluated without a hidden scan. After the bounded suffix replacement, the temporary buffer has length strictly less than $3C$. If it has length $\geq 2C$, push one full C-block; if it has length $< C$ and at least one physical block exists, pop one C-block and prepend it to the buffer; otherwise do nothing. This restores the unique canonical packing, including the case of zero blocks. A single stored block is one finite physical stack symbol; its push or pop is one real-time operation. □

Proof of the positive direction of Theorem 2.1. Define $\mathrm{NF}(U, V)$ by the axis, rank-zero, horizontal, vertical and northeast clauses. The original source tags, their bottom-to-top order, current residues, component and mode are explicitly recoverable from the combined control and labelled logical word. This proves full-source injectivity and history independence. Lemma 4.4 provides an exact finite top-local transition on every reachable source state. Lemma 4.5 turns it into a physical deterministic real-time transition reading the same input letter. Illegal pops, wrong tags, underflows and post-step mask violations are rejected without modifying the physical state. At the root the normal form is the initial physical state. Induction on guarded input length gives

$$\Delta_a(\mathrm{NF}(U, V)) = \mathrm{NF}(\delta_a(U, V))$$

for every defined input action a, together with exact reflection of undefined actions and terminal empty-stack acceptance. □

5 An effective signed-winding test and examples

Proposition 5.1 (Explicit-mask decision complexity)**.** *For an explicitly supplied $p \times q$ mask, the signed-winding condition is computable using $O(pq)$ word-RAM operations and $O(pq)$ words of memory. The coordinates manipulated have $O(\log(pq + 2))$ bits; a conservative schoolbook bit-operation upper bound is $O(pq \log^2(pq + 2))$.*

Proof. Process each vertex and each of its at most four signed arcs using a spanning-forest search. Every non-tree arc contributes a chord voltage. In each component, save the first nonzero voltage v. Another voltage with nonzero determinant with v certifies rank two; otherwise all nonzero voltages are collinear and their common sign type decides the rank-one condition. Each edge is processed constantly many times and each determinant costs a constant number of word operations. The potential magnitude is $O(pq)$ by the tree length bound. □

The explicit-mask graph criterion does *not* imply a small compiled machine. With a fixed logical alphabet of g cells and the bound C, physical stack symbols may be C-tuples, so a coarse physical alphabet bound is $1 + g^C$; the controller has similarly large, but finite, bounded-tag caches.

Example 5.2 (A non-affine periodic example)**.** *For periods $(p, q) = (4, 4)$, define the mask R by integer bitset 6599, where bit $i + 4j$ indicates membership of residue (i, j). Its row patterns at $j = 0, 1, 2, 3$ are respectively 1110, 0011, 1001, and 1000 in increasing i order. An exhaustive check of all $4^3 = 64$ coefficient triples shows that it is not representable as an affine-complement condition $ax + by + c \not\equiv 0 \pmod 4$, yet its graph satisfies the positive signed-winding condition and has a northeast rank-one component. Thus the example is genuinely outside the affine-complement modular family. The mask is explicitly specified here, and its signed residue graph can be verified directly from the definition.*

For the important special case $R = \{(x, y) : ax + by \not\equiv 0 \pmod{m}\}$, the coefficient condition is $a = 0$ or $b = 0$ or $a + b = 0 \pmod{m}$. This coefficient condition follows from the general signed-winding criterion and is included here as an explicit arithmetic specialization. The present paper requires no primality assumption on any period.

5.1 A sharp island bound and a physical-information lower bound

Proposition 5.3 (Uniform sharpness for rank-zero masks)**.** *For every integer $m \geq 1$ there exists a rank-zero mask with precisely m allowed residues and a root-reachable point satisfying* $\min(x, y) = m$. *Thus $|R|$ is the optimal universal bound in Lemma 4.1, viewed as a function of $|R|$ alone.*

Proof. Take $p = 2$, $q = m + 1$ and $R_m = \{(1, j) : 1 \leq j \leq m\}$. Every connected interior lift is a vertical chain of m points, so every voltage vanishes. Choose an odd $x_m \geq m$, move along the free X axis from the origin to $(x_m, 0)$, and push Y successively to height m. All intervening positive points have permitted residues and the endpoint has $\min(x_m, m) = m = |R_m|$. □

Proposition 5.4 (Source-information capacity; elementary counting bound)**.** *Suppose that, at a root-reachable height (x, y), all $s^x t^y$ assignments of independent source tags are reachable, where $s = |T_X|$, $t = |T_Y|$. Any full-source-injective encoding with at most Q finite-control states and a physical stack alphabet of $g \geq 2$ symbols must attain stack depth at least*

$$H(x, y) \;\geq\; \log_g \left(1 + \frac{(g-1)s^x t^y}{Q}\right) - 1$$

for some configuration at that height. In particular, when $s, t \geq 2$ and a legal family has $x, y = \Theta(n)$, every such representation has worst-case depth $\Omega(n)$ for fixed Q, g.

Proof. The $s^x t^y$ distinct source pairs require distinct physical configurations. There are at most $Q \sum_{k=0}^{H} g^k = Q(g^{H+1} - 1)/(g - 1)$ physical configurations with stack height at most H. Rearranging gives the claimed inequality. This standard information-counting argument is included only to quantify the physical meaning of full source preservation; it is not claimed as a novel combinatorial method. □

Finite checks. Independent implementations have tested signed-voltage classification, tagged-source transitions, and physical block thresholds for representative periodic masks. These finite checks are supplementary consistency evidence; the all-period assertions depend on the symbolic proofs in the preceding sections.

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Data availability. This is a theoretical study; no external dataset was used.
Code availability. Independent finite audit programs and summary results were used for internal verification. Research audit scripts and summaries are maintained as supplementary material; no public repository is asserted.
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